AN INTERNSHIP REPORT

ON

DecodeLabs Virtual Internship – Frontend Development

Submitted to

IILM UNIVERSITY
Greater Noida

In partial fulfilment of the requirements for the degree of

Bachelor of Technology (B.Tech)
Computer Science and Engineering (CSE) – Core

Submitted by

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Roll Number / Student ID: 25SCS1003002485

Batch: 2025–2029

Internship Organization	DecodeLabs
Internship Period	05/07/2026 – 05/08/2026
Certificate Issue Date	07/08/2026

CANDIDATE'S DECLARATION

I, **Unnayan Tyagi**, hereby declare that the internship report titled **“DecodeLabs Virtual Internship – Frontend Development”** has been completed by me as part of the academic requirements for the Bachelor of Technology (Computer Science and Engineering – Core) programme at IILM University, Greater Noida.

The report presents my learning and practical work during the DecodeLabs Virtual Internship from 05/07/2026 to 05/08/2026. The report has been prepared using the internship project materials supplied for the Frontend Development track and the completion certificate issued by DecodeLabs.

I further declare that the work described in this report represents my internship learning across the three assigned frontend development projects: Static Webpage Design, Responsive Web Layout, and Interactive Web Elements. I have not knowingly included confidential or fabricated project evidence in this report.

Student Name	Unnayan Tyagi
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Batch	2025–2029
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Date	______________________________

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to everyone who supported me during my DecodeLabs Virtual Internship in Frontend Development. The internship provided an opportunity to move beyond classroom concepts and practise the fundamentals of building web interfaces through structured projects.

I am thankful to **DecodeLabs** for providing the Frontend Development training material and project-based learning structure. The three projects helped me progress from basic HTML and CSS structure, to responsive design, and finally to JavaScript-based interaction and DOM manipulation.

I also express my gratitude to **IILM University** and the faculty members of the Computer Science and Engineering programme for their academic guidance and encouragement. The internship experience helped me understand how frontend concepts are connected when building a usable web interface.

Finally, I am grateful to my family and everyone who encouraged me throughout the internship period and helped me remain consistent in completing the assigned projects.

Student Name: Unnayan Tyagi

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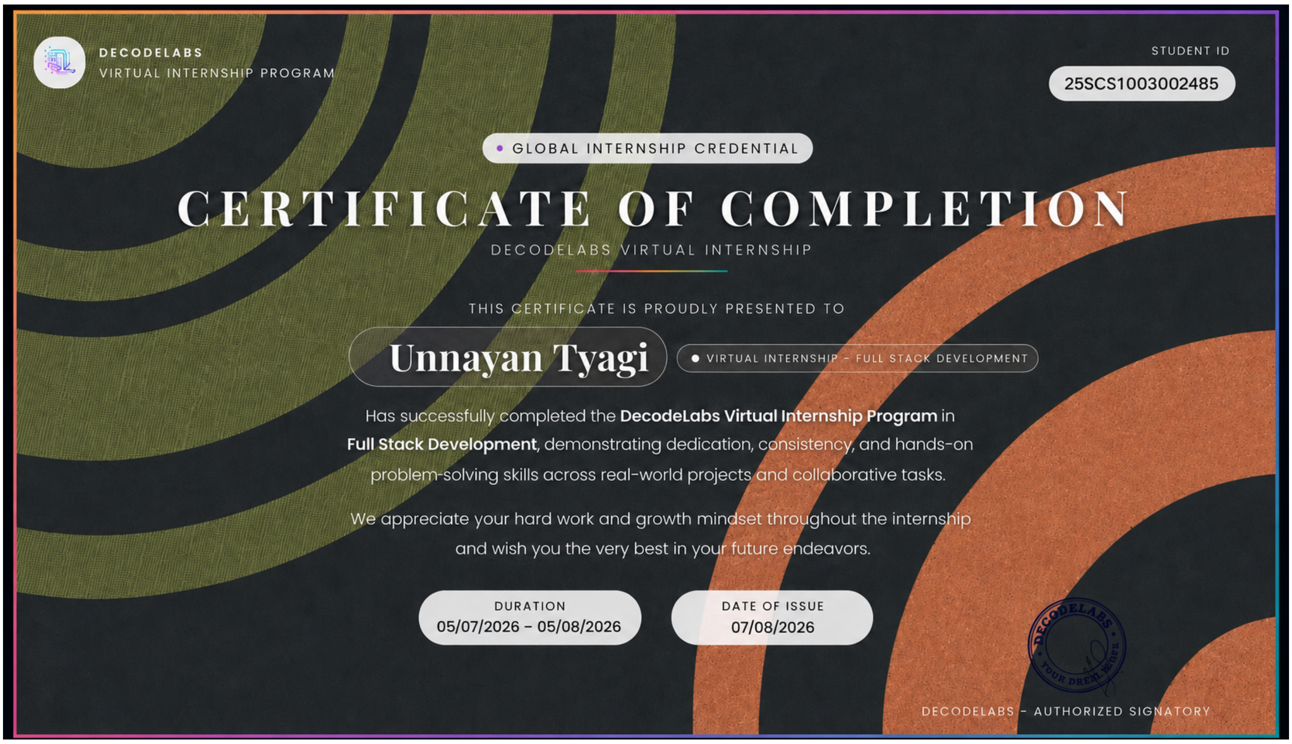
Note: The original supporting-document sections for an offer letter and communication proof have intentionally been omitted. The completion certificate supplied with the internship materials is included as the supporting credential.

3. INTERNSHIP COMPLETION CERTIFICATE



The above certificate is the DecodeLabs Global Internship Credential supplied for this report. It records the student ID 25SCS1003002485, the internship duration from 05/07/2026 to 05/08/2026, and the issue date 07/08/2026.

4. PROJECT DESCRIPTION

4.1 Introduction

This report presents the work completed during the DecodeLabs Virtual Internship in **Frontend Development**. The internship was carried out from 05 July 2026 to 05 August 2026. The supplied DecodeLabs training material is organised around three progressive projects, beginning with the structure and styling of a static webpage, moving to responsive layouts, and ending with interactive web elements using JavaScript.

The training sequence is designed around increasing frontend responsibility. Project 1 focuses on semantic HTML and CSS styling; Project 2 focuses on making a webpage adapt across different screen sizes using responsive techniques; and Project 3 focuses on connecting the interface to user actions through JavaScript and DOM manipulation.

4.2 Organization Profile

DecodeLabs is the organization identified on the supplied internship certificate and training materials. The internship material is presented as an **Industrial Training Kit** for Frontend Development and provides project-based learning tasks. The supplied material specifically covers the three projects documented in this report and describes the expected goals, requirements and skills for each project.

For this report, the organization profile is therefore described from the perspective of the internship programme itself: a structured virtual learning experience in which frontend concepts are introduced progressively through practical project requirements.

4.3 Problem Statement

Learning frontend development only through individual syntax examples can make it difficult to understand how structure, styling, responsiveness and interaction work together. A webpage may look correct on one screen size but fail on another, or it may have a visually complete interface without responding correctly to user actions.

The internship addresses this learning problem through a staged practical approach. The first project establishes a clean HTML and CSS foundation. The second introduces responsive behaviour so that the same interface can adapt to different screen sizes. The third adds JavaScript-based interaction and dynamic content updates, completing the progression from static structure to interactive frontend behaviour.

4.4 Project Objectives

- To understand the fundamentals of structuring a webpage with HTML and styling it with CSS.
- To practise semantic page organisation using headings, sections and images.
- To develop responsive webpages that work across different screen sizes.
- To understand and apply CSS media queries and responsive navigation.
- To improve spacing, alignment and layout techniques using modern CSS approaches.
- To add basic user interaction using JavaScript.
- To use buttons, toggles and dynamic content updates in a webpage.
- To understand the relationship between JavaScript and the Document Object Model (DOM).
- To complete all three assigned Frontend Development projects during the internship.

4.5 Scope of the Internship

The scope of the internship covers the core frontend concepts represented by the three supplied DecodeLabs projects. It includes webpage structure, visual styling, responsive behaviour and basic client-side interaction. The projects form a progression rather than three unrelated tasks: the skills introduced in one stage provide the foundation for the next.

The internship scope is focused on frontend development and the supplied project requirements. The report does not claim production deployment, organizational software development, or any technology that is not present in the supplied training material.

4.6 Technologies and Tools Used

Technology / Tool	Use during the internship
HTML	Page structure, headings, sections, images and semantic organisation
CSS	Styling, spacing, alignment and visual layout
CSS Media Queries	Adapting layouts for different screen sizes
Responsive Navigation	Navigation behaviour for smaller screens
CSS Grid	Two-dimensional page and macro-layout organisation
Flexbox	One-dimensional component and micro-layout organisation
JavaScript	Client-side interaction and behaviour
DOM Manipulation	Updating webpage elements dynamically in response to actions

4.7 Project 1 – Static Webpage Design

Project 1 is the structural phase of the Frontend Development track. The supplied material defines its goal as creating a static webpage using HTML and CSS. Its key requirements are proper HTML structure, use of headings, sections and images, and a clean and readable layout. The stated skills are HTML fundamentals and CSS styling.

Project Goal

The goal is to create a static webpage whose content is organised clearly and whose presentation is controlled through CSS. This stage establishes the relationship between markup and visual style before responsive and interactive behaviour is introduced.

Key Requirements

- Use a proper HTML structure.
- Use headings, sections and images appropriately.
- Maintain a clean and readable layout.
- Apply CSS to control the visual presentation of the page.

Learning from Project 1

This project develops the foundation needed for later frontend work. A well-structured page makes it easier to apply responsive rules and to target elements through JavaScript. The project therefore emphasises semantic integrity and a clear separation between content structure and styling.

Project 1 Outcome

On completion, the intern demonstrates an understanding of basic HTML page construction and CSS styling. The project provides the base interface on which responsive and interactive features can later be built.

4.8 Project 2 – Responsive Web Layout

Project 2 is described in the supplied material as the adaptability phase: **Responsive Web Layout**. Its goal is to build a webpage that works across different screen sizes. The key requirements include CSS media queries, responsive navigation, and proper spacing and alignment.

Core Concepts

- CSS media queries for adapting the layout at different viewport sizes.
- Responsive navigation so that navigation remains usable on smaller screens.
- Proper spacing and alignment across layouts.
- Responsive design and CSS layout techniques.
- Mobile-first thinking and fluid layout behaviour.

Layout Techniques

The supplied Project 2 material specifically discusses Grid and Flexbox as complementary layout techniques. Grid is suited to larger page-level or two-dimensional arrangements, while Flexbox can be used for one-dimensional component arrangements. The material also presents responsive navigation, accessible touch targets and zoom behaviour as considerations for mobile interfaces.

Project 2 Outcome

The project strengthens the ability to design interfaces that remain usable beyond a single desktop screen size. It introduces the idea that layout should respond to available space instead of being built around fixed dimensions.

4.9 Project 3 – Interactive Web Elements

Project 3 is the engagement phase of the supplied Frontend Development track. Its goal is to add interactivity to a webpage using JavaScript. The key requirements are buttons or toggles, basic user interaction and dynamic content updates. The stated skills are JavaScript basics and DOM manipulation.

Core Concepts

- JavaScript event-driven behaviour.
- Buttons and toggles that respond to user actions.
- Updating webpage content or visual state dynamically.
- Selecting and manipulating elements through the DOM.
- Keeping behaviour and presentation logically separated.

Interactive State

The supplied Project 3 material uses examples such as a dark-mode toggle to illustrate the flow from user input to JavaScript processing and then to a visible interface change. It also discusses dynamic classes, where JavaScript changes state through class manipulation while CSS controls the visual result.

Project 3 Outcome

This project completes the progression from static content to an interface that reacts to the user. It gives practical exposure to the connection between HTML elements, CSS presentation and JavaScript logic.

4.10 Methodology and Learning Progression

The internship followed a project-based progression using the three supplied Frontend Development projects. Each stage introduced a new layer of frontend capability and built on the previous stage.

Stage	Project	Focus
1	Static Webpage Design	HTML structure and CSS styling
2	Responsive Web Layout	Media queries, responsive navigation, Grid/Flexbox and layout adaptation
3	Interactive Web Elements	JavaScript, DOM manipulation, buttons/toggles and dynamic updates

4.11 Expected Outcomes

- Ability to create a structured webpage using HTML and CSS.
- Improved understanding of semantic page organisation and readable layouts.
- Ability to apply responsive design principles for different screen sizes.
- Familiarity with media queries, responsive navigation, CSS Grid and Flexbox.
- Ability to add basic interactive behaviour using JavaScript.
- Understanding of DOM manipulation and dynamic interface updates.
- A clearer understanding of how structure, styling, responsiveness and interaction work together in frontend development.

4.12 Conclusion

The DecodeLabs Frontend Development internship provided a practical progression through three important layers of web development. Project 1 established the HTML and CSS foundation, Project 2 extended that foundation to responsive layouts, and Project 3 introduced JavaScript-driven interaction. Together, the projects helped connect individual frontend concepts into a complete learning path.

The internship also reinforced the importance of clean structure, adaptable layouts and user-focused interaction. Completing all three projects provided hands-on exposure to the core concepts represented in the supplied DecodeLabs training material and contributed to a stronger foundation for further frontend development work.

5. BIBLIOGRAPHY / REFERENCES

The following sources were used as the basis for the internship report. The project descriptions and terminology have been kept aligned with the supplied DecodeLabs training material.

1. DecodeLabs, “Frontend Development – Project 1: Industrial Training Kit,” supplied internship training material, 2026.

2. DecodeLabs, “Frontend Development – Project 2: Industrial Training Kit,” supplied internship training material, 2026.

3. DecodeLabs, “Frontend Development – Project 3: Industrial Training Kit,” supplied internship training material, 2026.

4. DecodeLabs, “Global Internship Credential – Certificate of Completion,” issued to Unnayan Tyagi, Student ID 25SCS1003002485, 07/08/2026.

Supporting material intentionally omitted: Internship Offer Letter and Communication Proof.

— END OF REPORT —